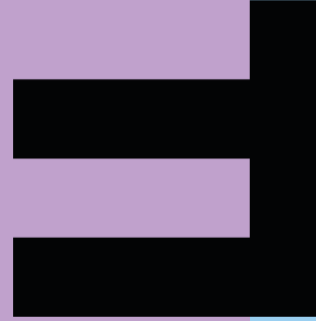


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Schmitt-Trigger

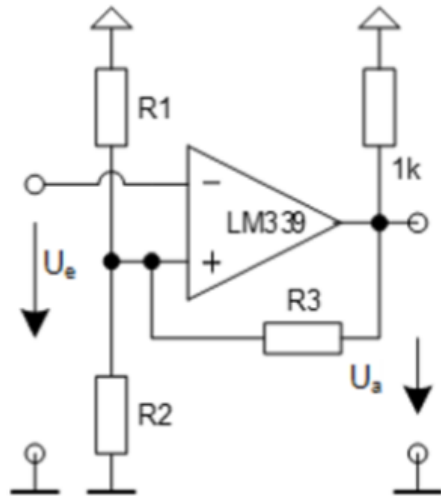
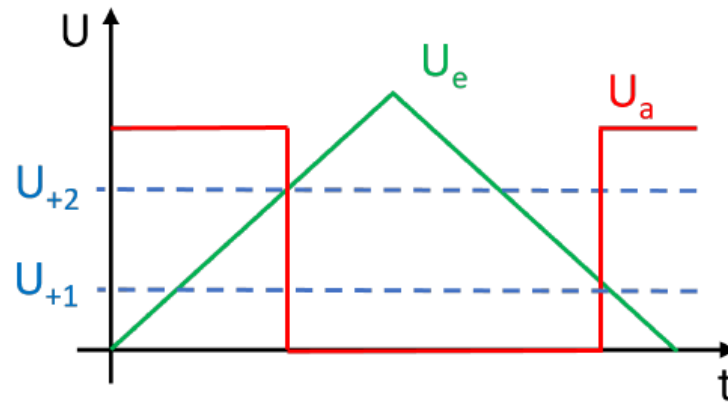
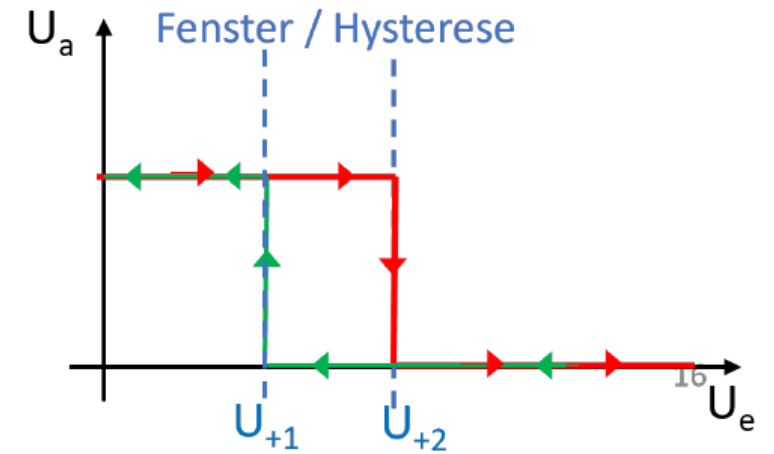


Abbildung: Schaltung



(a) über Zeit



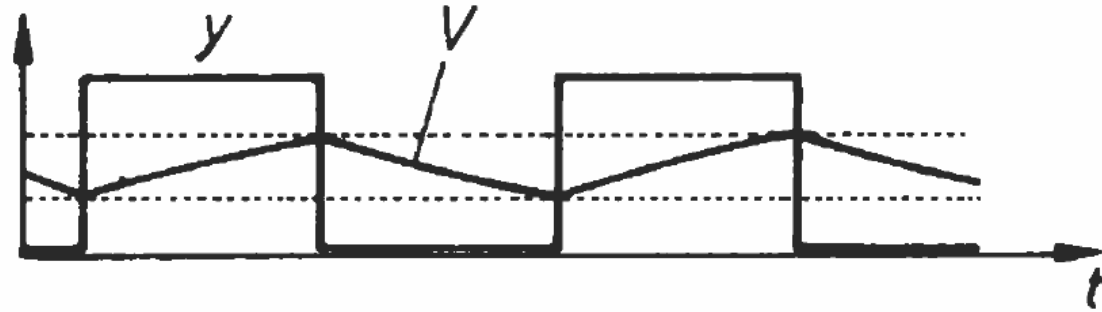
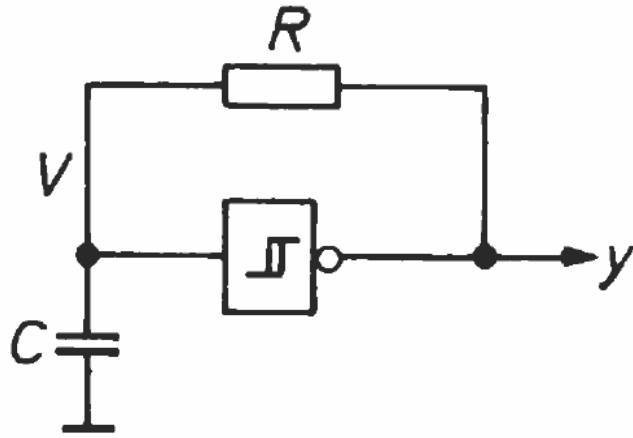
(b) Über Eingang

Abbildung: Signale

[Stefan Stark – Grundlagen der Elektronik, WS 2024]



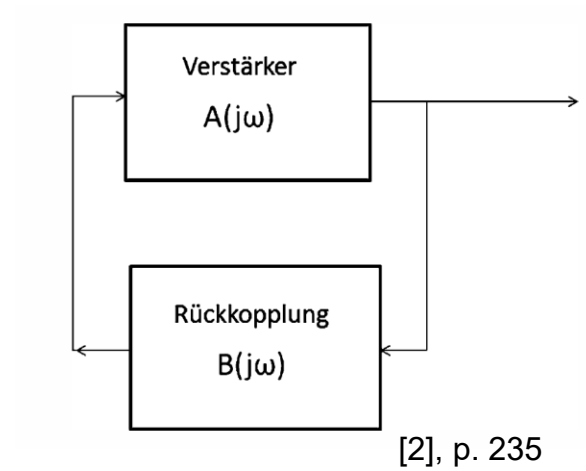
Schmitt-Trigger Oscillator



[1], p. 609



Barkhausen Criterion



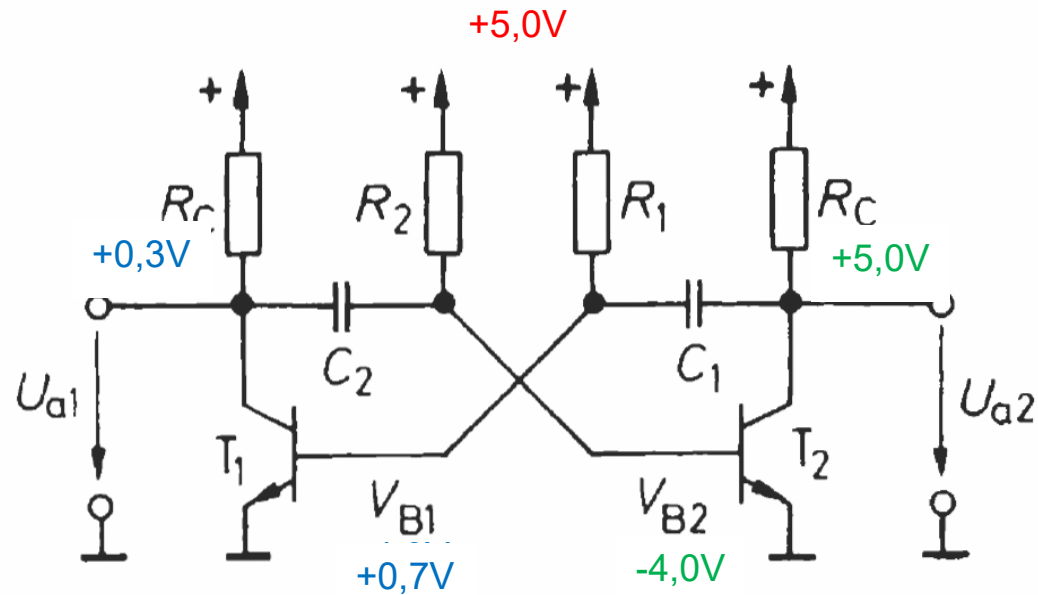
$$|A(j\omega)| \cdot |B(j\omega)| = a(\omega) \cdot b(\omega) = 1$$

$$\alpha(\omega) + \beta(\omega) = n \cdot 2\pi$$



Astable Multivibrator

- Is another name for oscillator

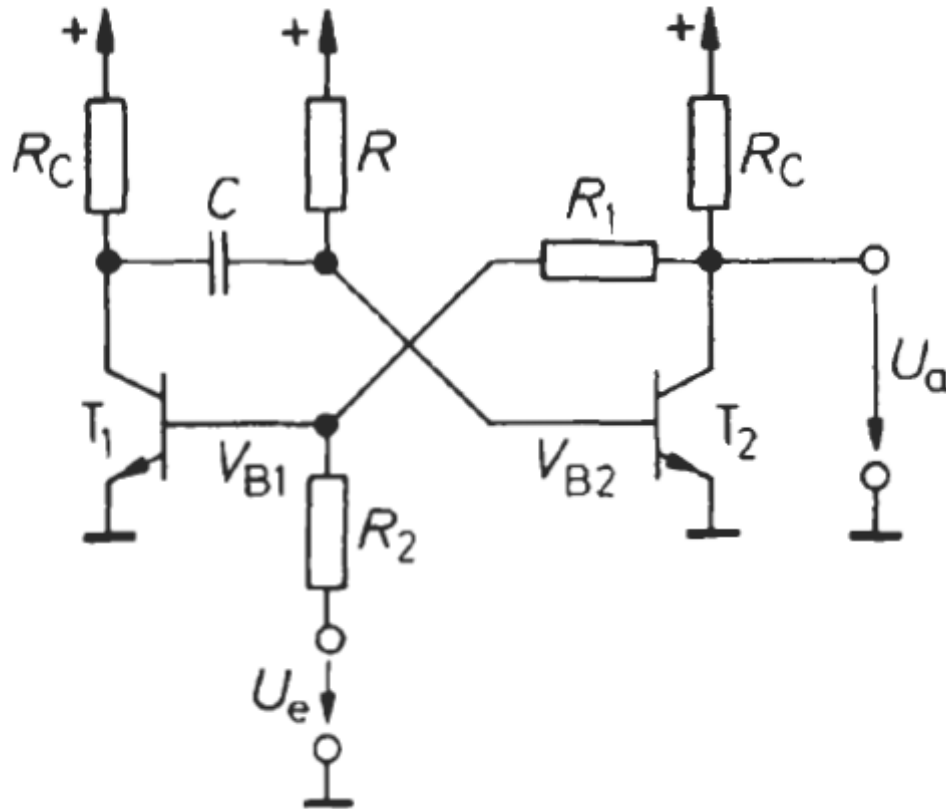


[1], p. 603



Monostable Multivibrator

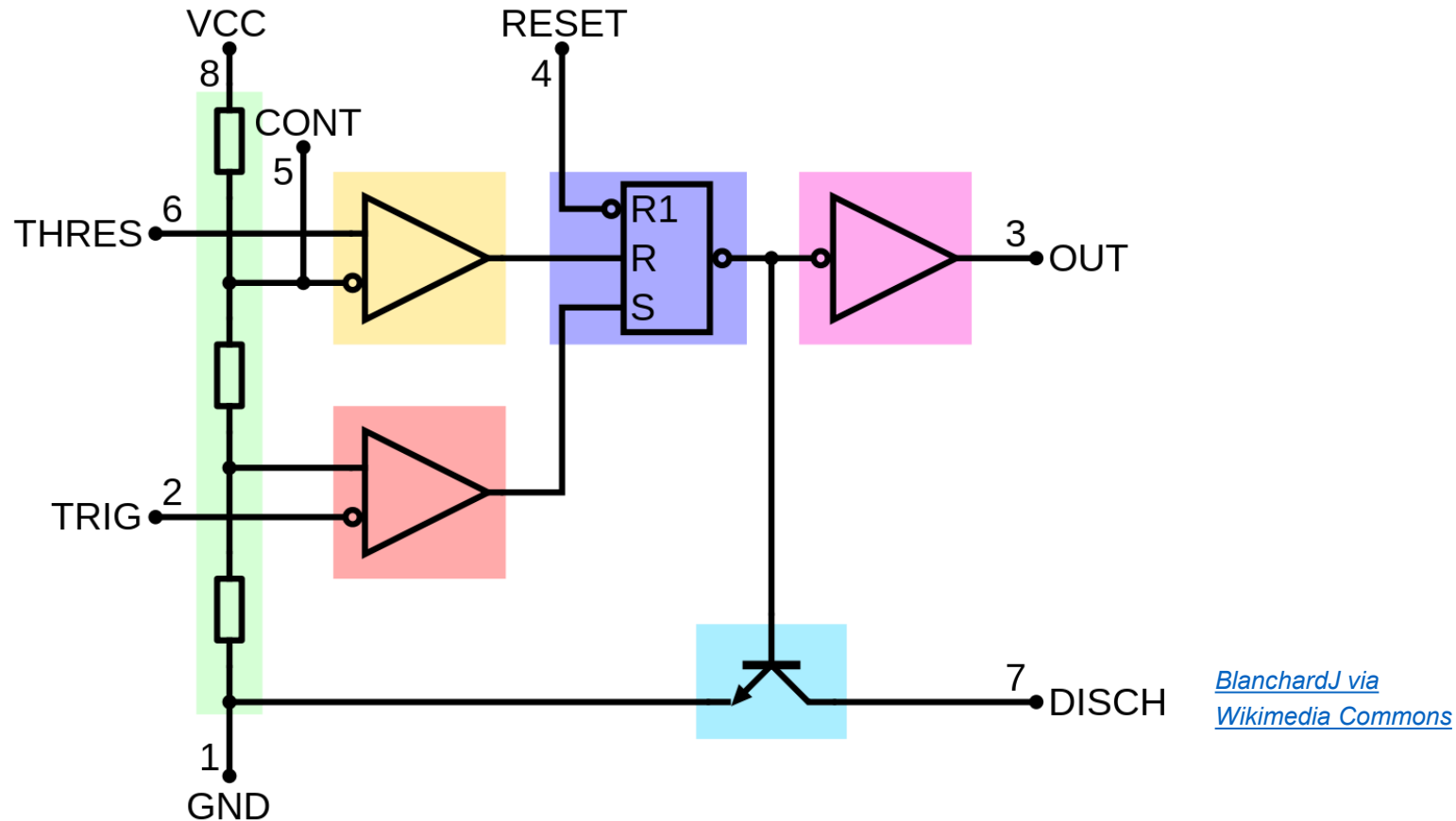
- By applying an external pulse, the status changes for a period of time



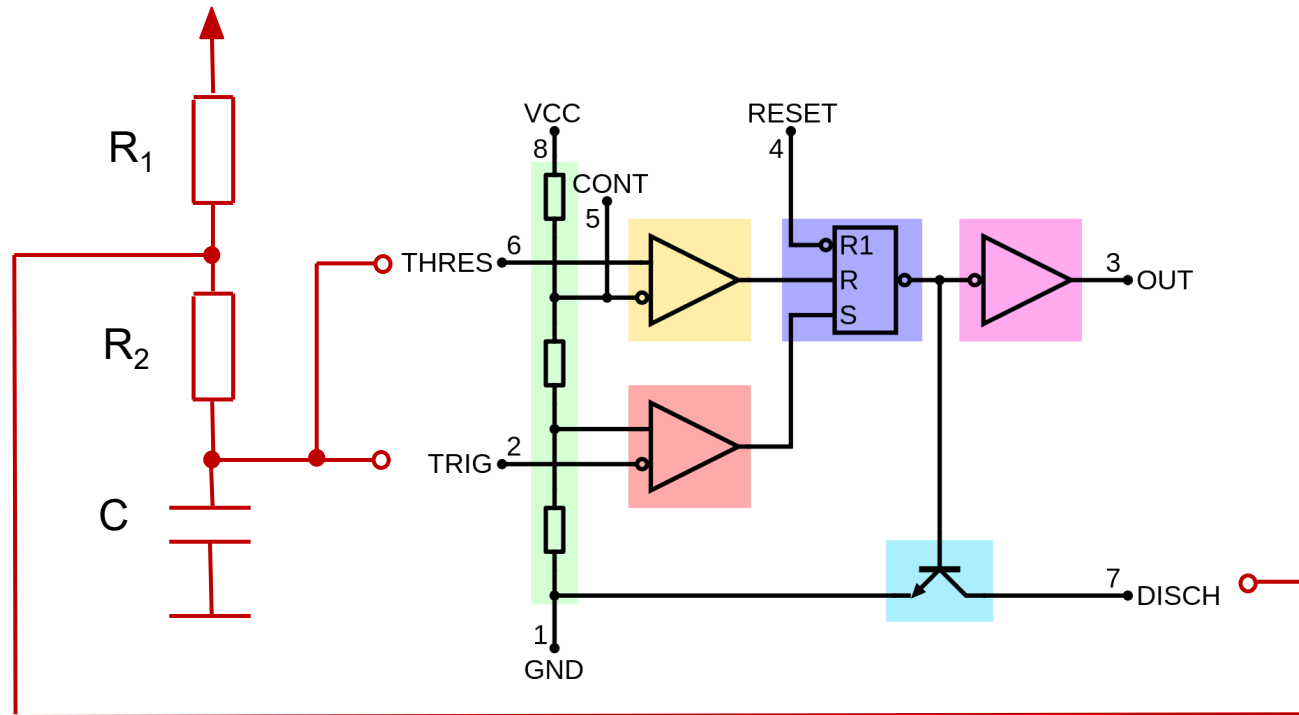
[1], p. 602



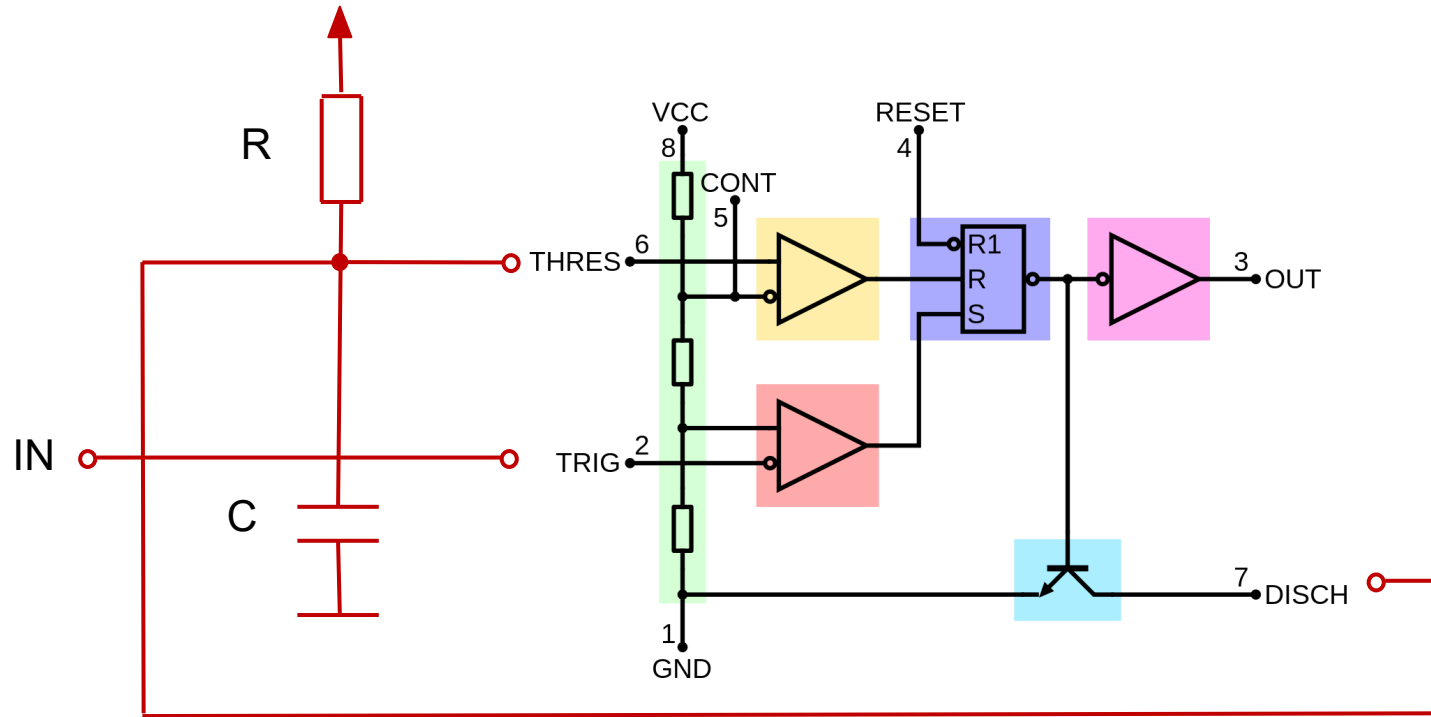
The NE555



Astable Multivibrator using NE555

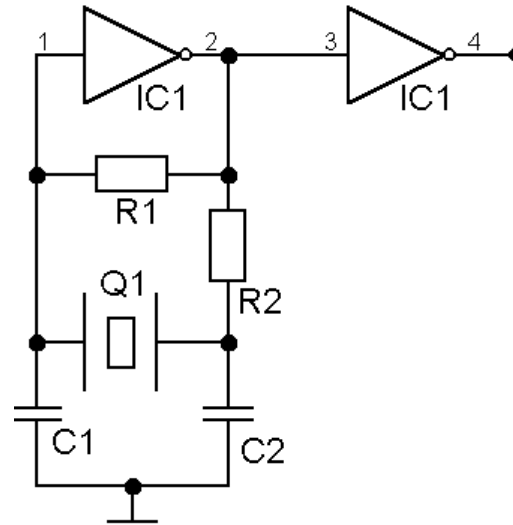


Monostable Multivibrator using NE555



Quartz Oscillator

- Typical quartz oscillator circuit (Pierce):



R_1 : 100 k Ω ... 10 M Ω

R_2 : 10 Ω ... 4,7 k Ω

C_1, C_2 : 10 pF ... 82 pF



Oscillator Parameter

Important oscillator parameter:

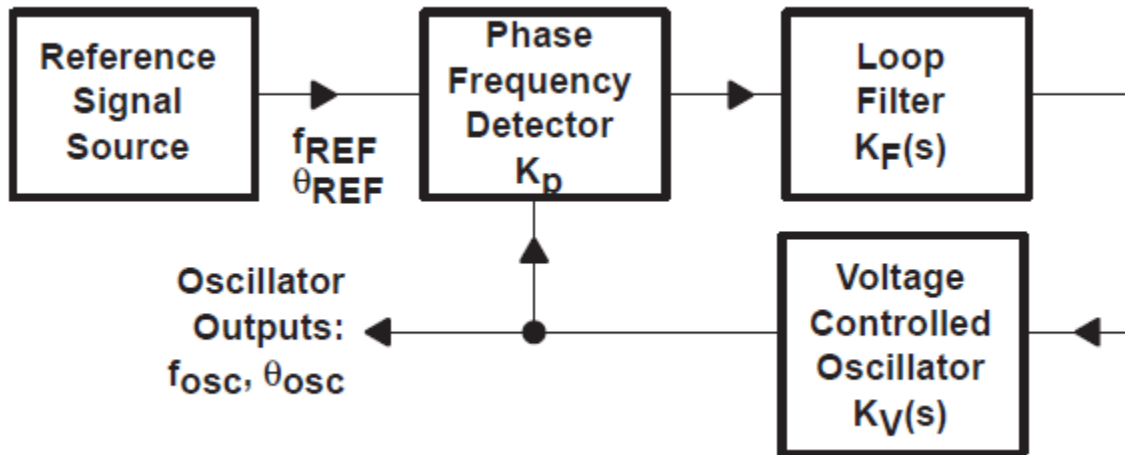
- Output waveform: sine (LC-oscillators), square, pulse, ...
- Output frequency
- Frequency tolerance
- Temperature stability
- Aging

Some typical values:

Parameter	Quartz crystal	Capacitor
Initial tolerance	20 ppm	5 %
Temperature drift	0.05 ppm/K	0.1 %/K
Aging	5 ppm/year	2 %/year



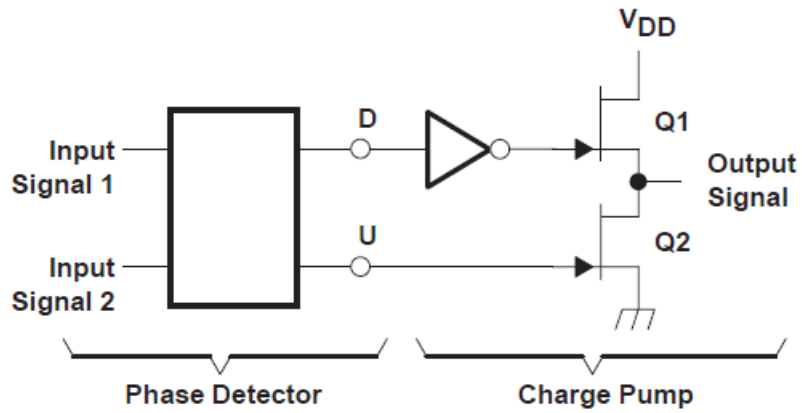
PLL (Phase Locked Loop)



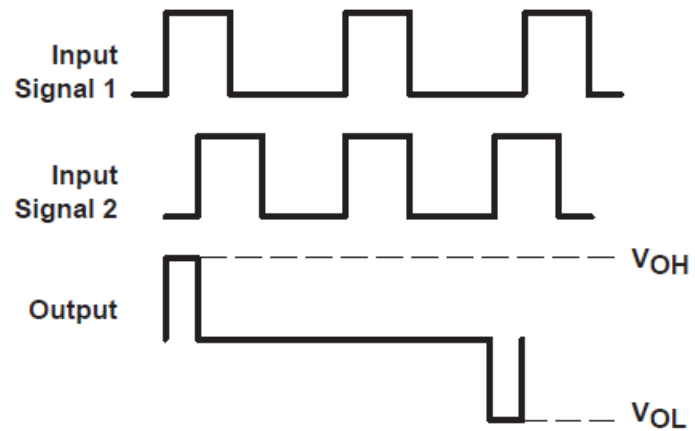
[3]



PLL - Detector



(a) 3-STATE PHASE FREQUENCY DETECTOR

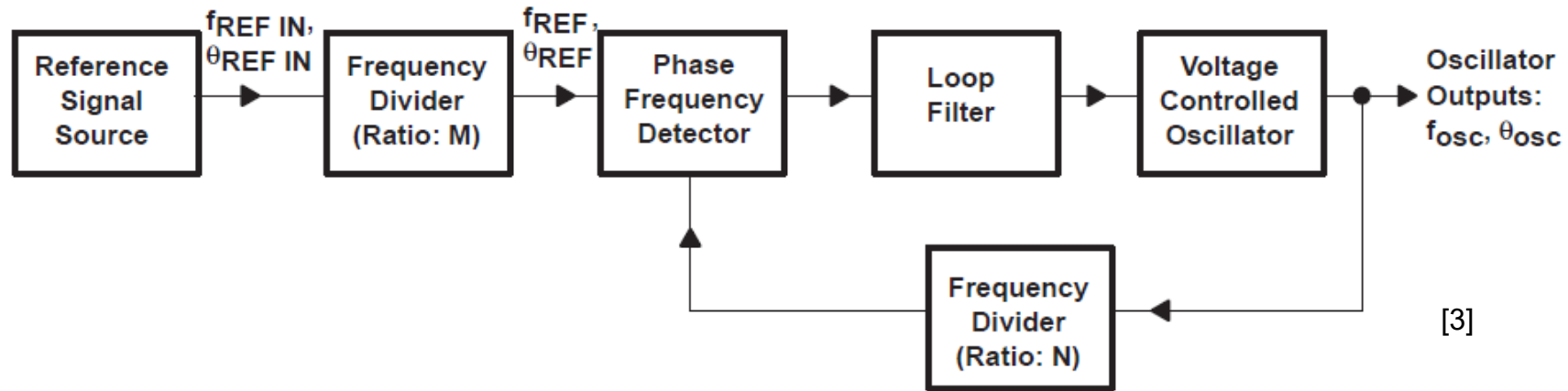


(b) PHASE FREQUENCY DETECTOR
INPUT AND OUTPUT SIGNALS

[3]

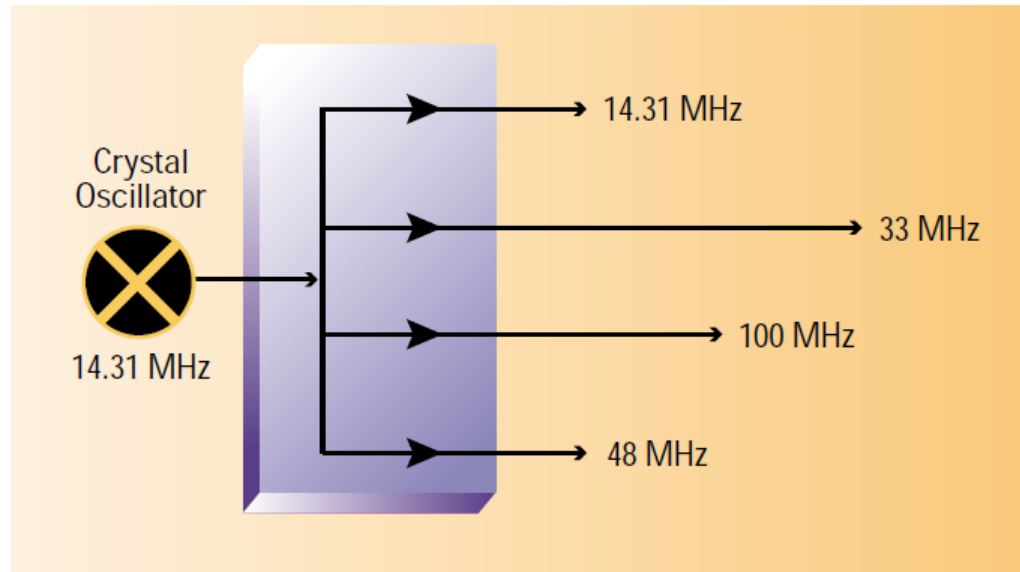


PLL – Application I



PLL – Application II

Clock Synthesizers/Drivers



[4]



References

- [1] Tietze, Ulrich; Schenk, Christoph (1999): Halbleiterschaltungstechnik. 11. Auflage. Berlin: Springer
- [2] Hering, Eckbert; Bressler, Klaus; Gutekunst, Jürgen (2014): Elektronik für Ingenieure und Naturwissenschaftler, 6. Auflage, Berlin, Heidelberg: Springer
- [3] „TLC2932 Application Report”, Texas Instruments, 1997, document SLLA011B,
<http://www.ti.com/lit/an/slaa011b/slaa011b.pdf>
- [4] Clock Driver Circuit Overview, www.ti.com



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